



Reverse Boundary Traversal Pattern

Master Notes (Pacific Atlantic + Surrounded Regions + Number of Enclaves)

Why These 3 Problems Are Studied Together

At first glance, these look like completely different problems.

Problem	Story
Pacific Atlantic	Water Flow
Surrounded Regions	O and X
Number of Enclaves	Land and Water

But internally they all ask the same question.

Can many cells reach the same destination?

Whenever you see this,

don't start DFS from every cell.

Instead think

```
Can I start from the destination  
and discover every cell that can reach it?
```

This is the **Reverse Boundary Traversal Pattern**.

Problem 1 — Pacific Atlantic Water Flow

Problem

Each cell has a height.

Water can flow

```
Higher
```

↓

Lower or Equal

Need to find

Cells

↓

Can reach Pacific

AND

Can reach Atlantic

Brute Force

For every cell

DFS

↓

Can Reach Pacific?

DFS

↓

Can Reach Atlantic?

Time

$O((n*m)^2)$

Optimization

Start from

Pacific Boundary

Reverse DFS

```
Neighbour >= Current
```

Mark

```
Pacific Reachable
```

Again

Start from

```
Atlantic Boundary
```

Reverse DFS.

Answer

```
Intersection
```

Meaning of visited

```
visited
```

```
↓
```

```
Can Reach Ocean
```

Problem 2 — Surrounded Regions

Problem

Replace every O that is completely surrounded by X.

Keep

```
Boundary 0
```

+

Connected 0

Brute Force

For every 0

DFS

↓

Can Reach Boundary?

Time

$O((n*m)^2)$

Optimization

Start DFS from

Boundary 0's

Mark

Safe 0

Finally

Every 0

AND

Not Safe

↓

Convert to X

Meaning of visited

visited

↓

Safe 0

Problem 3 — Number of Enclaves

Problem

Count land cells

1

that cannot reach boundary.

Brute Force

For every land

DFS

↓

Can Reach Boundary?

Time

$O((n*m)^2)$

Optimization

Start DFS from

Boundary Land

Mark

```
Escapable Land
```

Finally

```
Land
```

```
AND
```

```
Not Visited
```

```
↓
```

```
Count
```

Meaning of visited

```
visited
```

```
↓
```

```
Can Escape
```

Common Pattern

All three problems follow exactly the same thinking.

Step 1

Brute Force

```
For every cell
```

```
↓
```

```
Run DFS
```

```
↓
```

Can it reach destination?

Step 2

Observe

Repeated DFS

Many searches explore the same component.

Step 3

Reverse Thinking

Instead of

Cell

↓

Destination?

Ask

Destination

↓

Which cells
can reach me?

Step 4

Start DFS from destination.

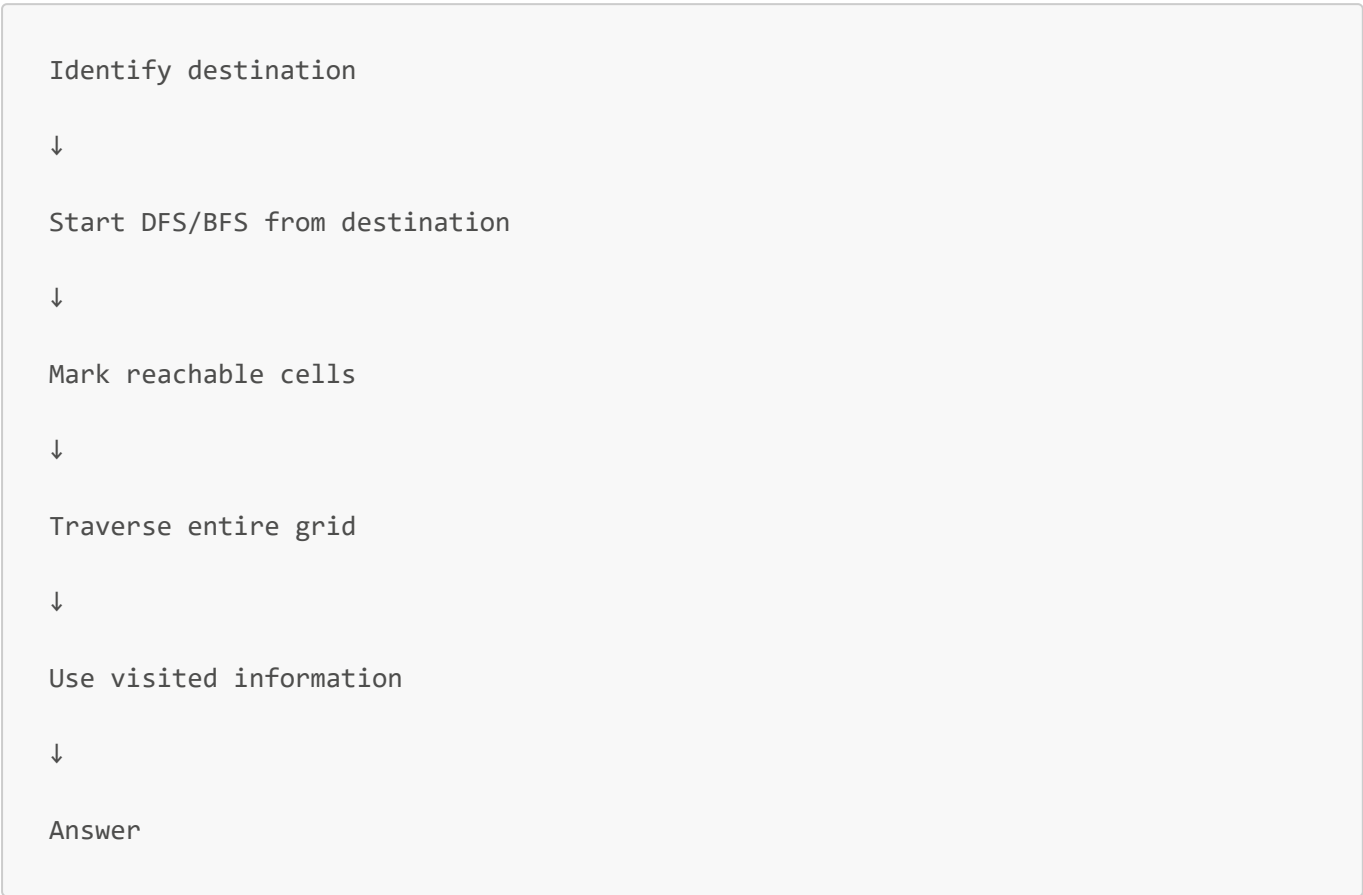
Step 5

Mark every reachable cell.

Step 6

Use visited information.

Generic Algorithm



Comparison Table

Feature	Pacific Atlantic	Surrounded Regions	Number of Enclaves
Brute Force Question	Can cell reach Pacific & Atlantic?	Can O reach boundary?	Can land reach boundary?
Destination	Pacific + Atlantic	Boundary O	Boundary Land
DFS Starts From	Ocean Boundaries	Boundary O's	Boundary Land
Reverse DFS Needed?	Yes	Yes	Yes
Movement Rule	>= (reverse flow)	Adjacent O	Adjacent Land
Visited Means	Can reach ocean	Safe O	Escapable Land
Final Step	Pacific ∩ Atlantic	Flip unsafe O to X	Count unvisited land
Output	Count	Modified Grid	Count

Recognition Keywords

Whenever you read words like

Reach
Flow
Boundary
Escape
Connected
Propagate
Eventually Reach
Can Reach
Ocean
Exit
Border

Immediately ask

Can I reverse the search?

Recognition Checklist

1.
Am I checking
every cell
individually?

↓

YES

↓

2.

Is every cell
trying to reach
the same destination?

↓

YES

↓

3.

Can I start
from destination?

↓

YES

↓

Reverse DFS/BFS

Mental Model

Never think

Cell

↓

Destination

Instead think

Destination

↓

Find Every Cell

One-Line Memory Trick

Many Cells

↓

One Destination

↓

Reverse Search

Pattern Template (Permanent Revision)

Problem asks

↓

Can every node

reach X?

↓

Brute Force

↓

DFS from every node

↓

Repeated work

↓

Reverse Thinking

↓

Start from X

↓

DFS/BFS

↓

Mark reachable

↓

Traverse grid

↓

Answer

Similar Interview Problems

Same Reverse Boundary DFS Pattern

- Pacific Atlantic Water Flow
- Surrounded Regions
- Number of Enclaves
- Closed Islands
- Flood Fill (variation)
- Eventual Safe States (graph version)

Same Multi-Source Traversal Family

- Rotting Oranges
- 01 Matrix
- Walls and Gates
- Shortest Bridge

- As Far From Land As Possible

Final Interview Takeaway

These problems are **not about water, O/X, or land**.

They all belong to the same underlying family:

When many cells independently ask "Can I reach the same destination?", reverse the search. Start from the destination (or destination boundary), traverse outward, mark all reachable cells, and derive the answer from the visited matrix.